Economic Analysis and Policy 78 (2023) 438–455

Contents lists available at ScienceDirect

Economic Analysis and Policy

journal homepage: www.elsevier.com/locate/eap

Modeling economic policy issues

Testing three views about the determinants of informal economy: New evidence at global level and by country groups using the CS-ARDL approach☆

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ARTICLE INFO

Article history:
Received 12 September 2022
Received in revised form 9 March 2023
Accepted 19 March 2023
Available online 27 March 2023

JEL classification:
J01
L83
C23
O57
C00
C01

Keywords:
Informal economy
Economic development
Institutional quality
Trade openness
CS-ARDL

ABSTRACT

This paper investigates the heterogeneous, long, and short-run and causal relationship between the size of the informal economy and three usual variables related to each of the most important theoretical approaches on informality, namely, the dualist, legalist, and structuralist view. Specifically, we consider GDP per capita, institutional quality and trade openness. We explore this relationship, using novel econometric techniques such as CS-ARDL panel data approach to analyses a sample of 133 countries over the period 1995 to 2017. The results show a long-run relationship between the variables at the global level and by country group. GDP per capita, institutional quality and trade openness are negatively associated with the size of the informal economy. The signs confirm the dual and legal hypothesis; however, the sign is opposite to that suggested by the structuralist hypothesis. Additionally, we obtain a U-shaped relationship between the size of the informal economy and the level of economic development. This means that the informal economy does not tend to disappear in the long run. In terms of policy implications, we provide support about the effectiveness of GDP per capita, institutional quality and trade openness in diminishing the size of the informal economy. Nevertheless, based on our results of a U-shaped relationship between informality and GDP per capita, public policy should be designed conditional on the level of development of a certain country.

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1. Introduction

The informal economy is a global phenomenon and its prevalence has been increasing in the recent decades (La Porta and Shleifer, 2014; Chen, 2006; Gerxhani, 2004; Schneider and Buehn, 2013; Schneider, 2017; Elbahnasawy, 2021; Dell'Anno and Solomon, 2008). For example, the informal sector comprises a large share of employment in developing countries (Dell'Anno and Solomon, 2008; Aktaruzzaman and Farooq, 2020) and a relevant percentage of economic

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https://doi.org/10.1016/j.eap.2023.03.016
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transactions at global level (Medina and Schneider, 2018). Besides, the size of the informal economy varies considerably across countries at different levels of development (La Porta and Shleifer, 2014; Loayza, 2016a).

Research about the informal economy has generated a fruitful debate, starting with the work developed by Hart (1973) during the 1970s, who was the first to coin the term informality. During the last decades, three views have dominated the debate on the informal economy (Chen, 2005, 2006; De Mel et al., 2013; Adom, 2016; Williams and Bezeredi, 2018; Dell'Anno and Solomon, 2008). The first corresponds to the dual view and is related to the dual theory of development (Lewis, 1954). From this perspective, the informal economy is a by-product of under-development (see Freije and Monteferrante, 2002; Banerjee and Duflo, 2005; La Porta and Shleifer, 2008, 2014). Likewise, previous studies find that much of the informal activities are merely for survival purposes, rarely grow and often supplement the income of poor households (Tokman, 1988; Nikopour and Shah Habibullah, 2010; La Porta and Shleifer, 2008, 2014). La Porta and Shleifer (2014) for instance, argue that the informal economy will decline sharply as countries reach higher levels of economic development.

The second perspective corresponds to the legal view and is related to institutional economics (North, 1990). Proponents of this view consider that informality is a consequence of the high transaction costs generated by a country's legal and institutional environment (De Soto, 1989; Feige, 1990; Perry and Maloney, 2007; Loayza, 2016a). De Soto (1989) argues that the presence of a deficient regulatory and legal framework increases the cost of staying formal and diminishes the likelihood of detection and prosecution. Various aspects of institutional quality, such as corruption (Friedman et al., 2000), excessive business regulations (Djankov et al., 2002; Enste, 2010; Ulyssea, 2010), labor market rigidities (Klapper et al., 2006; Araujo and de Souza, 2010; Loayza and Rigolini, 2011), high tax burden (Prado, 2011; Elgin, 2015), property rights (Acs et al., 2008; Salinas et al., 2020), and political stability (Baklouti and Boujelbene, 2020) have been noted as important determinants of the informality.

Finally, the structuralist view places the new logics of labor contracting and outsourcing of production, arising from the process of opening and deregulation of the world economy, as the most relevant determinants of the size of the informal economy (Castells and Portes, 1989; Chen, 2005). The structuralist view claim that trade liberalization is expected to increase competition for domestic producers and, in the face of this challenge, firms are given a greater incentive to cut costs, subcontracting to small informal firms some activities of the production process (Fiess and Fugazza, 2008). However, at the empirical level, the effects of trade openness on the informal economy have been ambiguous, and therefore are open for further research (Fiess and Fugazza, 2008; Birinci, 2013).

Besides the differences with regard to the determinants of informality, the three views differ in the way of explaining the connection between formal and informal sectors (Chen, 2005). The dual view of informality suggests that formal and informal workers coexist, operating in disconnected economic spaces, using different technologies, and producing distinct goods (La Porta and Shleifer, 2008, 2014; Ulyssea, 2010; Dell'Anno and Solomon, 2008). In contrast to the dual view, the structuralist view predicts that there is a symbiotic relationship between the formal and informal economy, where some production processes are outsourced to small informal enterprises (Jones et al., 2006). Particularly, the openness and deregulation of the world economy foster the creation of productive synergies between the two sectors to maintain market competitiveness (Piore and Sabel, 1984; Castells and Portes, 1989). Lastly, the legalistic view challenges the dual view and considers that informal firms are alike formal ones. According to this view, the existence of an excessive regulatory burden forces many entrepreneurs into informal economy (De Soto, 1989).

In this context, the prime contribution of this paper is to deepen the analysis by providing new empirical evidence on the validity of these three alternative and divergent theoretical approaches to informality, that had led to different lines of empirical research. In this sense, we use advanced second-generation econometric techniques, such as CS-ARDL, and perform short and long-run empirical analyses, taking in consideration the measures have been proposed to operationalize the concept of informal economy. We use the data presented by Medina and Schneider (2018), who estimate the size of the informal economy using a structural equation model. This measure includes all economic activities hidden from official authorities for monetary, regulatory, and institutional reasons (Medina and Schneider, 2018).

Additionally, as noted in the dual, legal, and structuralist literature, three usual measures are considered as independent variables: GDP per capita, institutional quality, and a measure of trade openness. For example, the dual view suggests that the size of the informal economy is inversely related to the level of economic development (La Porta and Shleifer, 2008). On the other hand, the legal view argues that better institutional quality is related to a lower level of informal activity (Friedman et al., 2000). Finally, our measure of trade openness attempts to capture the structuralist argument that relates the openness and deregulation of the world economy to a larger size of the informal economy (Castells and Portes, 1989).

In this paper we make four important contributions to the literature on informal economy. First, there have been no major empirical studies to examine divergent theoretical approaches on informality in an integrated manner, especially considering a long-run perspective. Such an analysis is very relevant, given the multidimensional nature of the informality phenomenon (Chen, 2006). Additionally, our approach of studying a possible long-run relationship between these variables and the size of the informal economy is necessary given the low variation of the informal economy in the short-run (Elgin and Erturk, 2016; Loayza, 2016b).

Secondly, most empirical studies on the informal economy use conventional econometric approaches that assume cross-sectional independence among panel units. However, these assumptions are difficult to satisfy given the economic, social and cultural interdependence that exists across countries, and therefore biases and erroneous conclusions are to be expected (Robertson and Sarafidis, 2015). To avoid these limitations, this study applies recently developed second-generation cointegration techniques such as CS-ARDL model (Chudik and Pesaran, 2015; Chudik et al., 2016). This approach

provides robust estimates in the presence of cross-sectional dependence (CSD), cross-country heterogeneity, potential simultaneity, and generates both long and short-run coefficients along with an error correction mechanism. In addition, the estimator also considers probable omitted variable bias (Chakraborty and Mazzanti, 2021).

Third, previous works have not looked for a possible heterogeneous effect of our independent variables on the size of the informal economy across levels of development. Specifically, we examine if each of the theories can adequately explain the emergence of informality in countries at different stages of development. For example, the structuralist vision can probably explain the dynamism of informal activity in Latin America countries, where industrial subcontracting is a central feature of informal activity (Yusuff, 2011). However, subcontracting is not an important feature of informal activity in African countries (La Porta and Shleifer, 2014). We attempt to fill this gap by splitting the sample into different subsamples according to the Atlas classification of year 2020 developed by the World Bank. The World bank divides countries into four groups: upper income countries (UIC); upper-middle income countries (UMIC), lower middle-income countries (LMIC), and low-income countries (LIC).

A last contribution of this article refers to the exploration of a potential non-linear long-run relationship between the size of informal economy and the level of economic development. Basically, there might be a tipping point in the relationship between the level of economic development and informal economy. Specifically, when GDP per capita moves beyond the tipping point, the informal economy tends to increase. To the best of our knowledge, it is among the first studies to explore this possible long-run nonlinearity.

In the process of assessing the heterogeneous long and short-run, and causal relationship between a measure of informal economy and three independent variables related to each of the main theoretical perspectives on informality, we use a large panel dataset of 133 countries over a relatively extensive period (1995 to 2017) to ensure the robustness of our analysis, both at the global and country group level. For this, we apply panel cointegration and causality tests. Specifically, after examining the existence of cross-sectional dependence and slope heterogeneity, we use the second-generation panel unit root test (Pesaran, 2007) to find the integrating properties of the variables and the panel cointegration test proposed by Westerlund (2007) for examining the long-run equilibrium between the variables.

Our empirical results confirm a long-run relationship between our variables of analysis at the global level and by country group. GDP per capita, institutional quality and trade openness are negatively associated with the size of the informal economy. The signs obtained in our analysis confirm the dual and legal hypothesis but contradict that suggested by the structuralist hypothesis. In general, these findings hold both in the short and long run, at the global level and by country group. However, for the case of low-income countries, we do not find a significant short-run relationship between institutional quality and the size of the informal economy. Interestingly, our results confirm the presence of a U-shaped relationship between the size of the informal economy and the level of economic development. This means that the informal economy does not tend to disappear in the long run.

Although many of these findings are not surprising, particularly those related to the dual and legal views, we make a clear methodological contribution derived from the use of co-integration techniques which is uncommon in the literature on informality. Indeed, to the best of our knowledge, we are the first researchers to use these techniques to investigate this relationship, which involve the following benefits. First, the use of this technique allows for a dynamic analysis that enables us to treat all variables as endogenous (Berdiev and Saunoris, 2016). Second, this dynamic analysis allows us to isolate the long-run from short-run effect, that had not been considered in previous studies. Besides, our approach also allows us to test the causal relationships between the variables. Finally, the technique allows us to perform the analysis among a large sample of upper; upper-middle; lower-middle, and low-income countries.

The rest of the paper is organized as follows. Section 2 reviews some of the existing literature on the subject. Section 3 presents the data used, Section 4 presents our methodology, and Section 5 is devoted to the empirical results and discussion, while Section 6 offers some conclusions.

2. Literature review

Three views are prevalent in the literature on the informal economy and its determinants: the dualist, legalist and structuralist (Chen, 2006, 2016; De Paula and Scheinkman, 2007). The first has its origins in the dual theory of development (Lewis, 1954). The second perspective is related to institutional economics (North, 1990). Finally, the third perspective considers the informal economy as a by-product of the deregulation and openness of the world economy (Castells and Portes, 1989; Chen, 2006, 2016).

The dual view of informality has its origins in the seminal work of Lewis (1954). According to Lewis' model, the informal or traditional sector will decrease as the modern sector develops, absorbing the jobs of the traditional sector. This view characterizes labor markets as segmented and dualistic (Lewis, 1954; Harris and Todaro, 1970; Fields, 2009). In Lewis's theoretical framework, the traditional or subsistence sector is related to the rural sector, which primarily develops in agriculture, while the modern sector develops in cities and is related to industry. Lewis' ideas were expanded by Harris and Todaro (1970). In this model, rural workers did not transit directly to employment in the modern sector, but to a traditional urban sector, while finding employment in the modern sector (Taiwo, 2013; Busso et al., 2021). According to this model, the urban traditional sector arises because of the inability of the urban economy to provide permanent jobs in the modern sector, and largely corresponds to what is currently called the informal economy.

From the dual view, the informal economy is a residual and transitory sector (Packard, 2007; Williams and Bezeredi, 2018) which consists primarily of self-employed workers who develop subsistence activities (see Gindling and Newhouse,

2012; Margolis, 2014). Additionally, they have a low level of job satisfaction (Aguilar et al., 2013) and absence of entrepreneurial traits (De Mel et al., 2008). Also, it is formed by small-scale firms that use low technology, little capital, and rarely grow and transition to formality (McKenzie and Woodruff, 2006; Mandelman and Montes-Rojas, 2009; Schoar, 2010; Monteiro and Assunção, 2012; De Andrade et al., 2013; De Mel et al., 2013; Diao et al., 2018). These microenterprises are operated by lower human capital microentrepreneurs (Tokman, 1988; Gennaioli et al., 2013). Moreover, informal firms generate low-productivity employment, while high-productivity jobs are created by formal firms (Bosch and Esteban-Pretel, 2012), whose productivity levels allow them to pay taxes and meet transaction costs arising from the regulatory framework (Gutiérrez-Romero, 2021). Therefore, the informal economy is associated primarily with poverty and social exclusion (Tokman, 1988; Gulyani and Talukdar, 2010; Banerjee and Duflo, 2011).

The dual perspective, presents the formal–informal sectors as two separate spheres, whose firms look very different, and which rarely interact (Nadin and Williams, 2012; Nordman et al., 2016). For example, micro-level studies confirm large productivity differences between formal and informal firms (Kathuria et al., 2013). Some papers find that this productivity gap is explained by differences in the educational level of managers (La Porta and Shleifer, 2008; De Paula and Scheinkman, 2011; Gennaioli et al., 2013). Likewise, there is also duality with respect to product quality. The highest quality good produced in the formal sector and the lowest quality produced in the informal sector (Banerji and Jain, 2007). Likewise, formal firms target mainly the middle classes, who are willing to pay a little more for the quality of the products (Banerjee and Duflo, 2008).

The dual view is related to modernization theories, which suggest that economic development occurs through different stages (Rosenstein-Rodan, 1943; Murphy et al., 1989; Rostow, 1990). Thus, the dual perspective sees the informal sector as transitory, and, therefore, its role will diminish as countries advance in their level of development (Bosch and Esteban-Pretel, 2012). However, this process of transition from informal to formal employment will be slow, and under certain conditions the adjustment will be achieved in the long run. Moreover, in this perspective, development does not come from the conversion of informal enterprises into formal ones, but from enterprises that started in the formal sector and are operated by individuals with better human capital and management skills (La Porta and Shleifer, 2014).

Thus, from the conceptual framework of the dual vision, we would expect that the higher the level of economic development, the smaller the size of the informal economy. Based on these findings, we propose the following hypotheses:

H1a: The higher the level of economic development the smaller the size of the informal economy both globally and by country group.

Additionally, the dual view suggests a simple linear and negative relationship between economic development and the size of informal economy. However, some authors have warned the potential existence of a nonlinear relationship in the long run (Wu and Schneider, 2019). The argument is that productivity improvement and technology advance can expand the size of informal economy by providing better salaries and as a result, people obtain financial freedom to an extent that they are willing to consider informal jobs (Wu and Schneider, 2019). This suggests a U-shaped relationship between the size of the informal economy and the level of economic development. Thus, we postulate the U-curve hypothesis:

H1b: There is a nonlinear relationship (U-shaped) between the level of economic development and the size of the informal economy

The second perspective, called the "legal view" is related to institutional economics (North, 1990; Feige, 1990). This perspective had a great impulse after the publication of the book The Other Path (De Soto, 1989). In this book, Hernando de Soto reveals the results of his fieldwork on the steps and time required to establish a business in his native Peru. The research revealed the existence of an excessive regulatory framework, which forced many entrepreneurs to enter or remain in the informal economy in order to avoid the costs, time and effort of formal registration (De Soto, 1989). In other words, the legal vision challenges the dual perspective, which sees the informal sector as residual and transitory. Under this perspective, the informal economy is a dynamic sector, a source of entrepreneurship and creativity, and whose entrepreneurs are constrained from crossing the line into formality by the high costs of complying with government regulations (Perry and Maloney, 2007; Araujo and Rodrigues, 2016).

Theoretical studies by Rauch (1991), Fortin et al. (1997), Johnson et al. (1998) and Loayza and Rigolini (2011) argue that excessive regulations end up causing labor-market segmentation and size dualism. These models infer that the high transaction costs imposed by the regulatory framework increase the opportunity costs of operating in the formal economy and push many small firms into informality and encourage many unskilled workers to go to the informal sector. Thus, from this approach, it is the institutional framework that determines the composition of the formal and informal sectors (Feige, 1990; Perry and Maloney, 2007).

For example, empirical evidence suggests that better institutional quality is related to a reduction in the size of the informal economy (Loayza and Rigolini, 2011; Hassan and Schneider, 2016; Saunoris and Sajny, 2017; Berdiev and Saunoris, 2018; Salinas et al., 2020). Basically, institutional quality affects the costs and benefits of operating in the formal economy vis-'a- vis the informal economy (De Soto, 1989; Perry and Maloney, 2007). For example, excessive regulations and very high registration costs deter many entrepreneurs from entering or moving to the formal sector of the economy (Djankov et al., 2002; Klapper et al., 2006; Loayza et al., 2009). Moreover, in countries with weak rule of law, firms are more likely to evade their fiscal and regulatory responsibilities, and choose to remain partially informal (Loayza and Rigolini, 2011; Ayyagari et al., 2007). Therefore, based on these results, we propose the following hypotheses:

H2: Better institutional quality is related to a smaller size of the informal economy both globally and by country group.

In contrast to the dual view, which suggests a disconnect between the informal and formal economies, the structuralist view points to the existence of important and complex linkages between the two sectors (Moser, 1978; Castells and Portes, 1989). These links were developed mainly from the process of restructuring the global economy which began during the 1970s and 1980s, as a consequence of a strong deregulation and opening of the world economy and the beginning of a new technological revolution, led by Information and Communication Technologies (ICTs) (Castells, 2004). Both events made it possible for many firms to transit from a vertically integrated mass production system to a flexible production system (Piore and Sabel, 1984), where parts of the production process can be subcontracted or outsourced to other formal firms or small informal enterprises (Jones et al., 2006), located in different parts of the world, and linked through global production networks (Coe and Yeung, 2015), or also called global value chain (Gereffi et al., 2005). In other words, from the structuralist view, these processes led the informal economy to become a subordinate and dependent sector of the modern capitalist sector (Castells and Portes, 1989).

At the empirical level, important links between the formal and informal economy have been noticed in the literature, through the use of subcontracting strategies by formal firms, many of which are part of global production networks (Carr et al., 2000; Laha, 2019; Strasser, 2015; Dannenberg et al., 2016; Das, 2017; Rogan et al., 2017; Vicol et al., 2019). For example, field studies conducted in several developing countries suggest that home-based workers and small informal firms form part of producer- and buyer-driven commodity chains through subcontracting relationships with formal firms (see Basole and Basu, 2011; Unni and Rani, 2008; Chen and Sinha, 2016; Rogan et al., 2017; Sial, 2020). These linkages are sustained, mostly from informal arrangements based on social relationships (Dannenberg et al., 2016).

Likewise, the process of opening and deregulation of the world economy implied the need for internationalization of domestic production systems (Coe and Yeung, 2015). For Castells and Portes (1989), outsourcing was the strategy used by many local firms to respond to the new competitive challenges imposed by the opening of economies. The objective of the strategy of subcontracting to small independent, informal firms was related to cut costs and increase flexibility (Chen, 2016). At the empirical level, relationships between formal and informal firms through subcontracting have been found in the manufacturing sector (Mukim, 2011; Ghani et al., 2013; Moreno-Monroy et al., 2014; Das, 2017), services (Henderson and Smith, 2009; Sial, 2020) and agriculture (von Oppenkowski, 2019; Meagher, 2019). Therefore, increases in outsourcing from the formal sector motivated by greater trade openness and consequently an intensified international competition would be associated with an increase in employment and activities in the informal economy. Thus, in countries with greater trade openness we expect a larger size of the informal economy.

H3: Greater trade openness is associated with a larger informal economy both globally and by country group.

3. Data

We use secondary datasets obtained from various sources, covering the timeline from 1995 to 2017, including 2926 observations, corresponding to 133 countries. The selection of countries was based on data availability. The countries were also divided into four subpanels according to the World Bank income classification: upper income countries (44 countries); upper-middle income countries (37 countries), lower middle-income countries (30 countries), and low-income countries (22 countries). The dependent variable of the study is the size of the informal economy. We use data from Medina and Schneider (2018), who estimate the size of the informal economy using indirect estimation methods. Specifically, they use a special type of modeling based on structural equations. The measure includes all economic activities that are hidden from official authorities for monetary, regulatory and institutional reasons (Medina and Schneider, 2018). The database has been widely used in the literature on informality (see Colombo et al., 2019; Baklouti and Boujelbene, 2020; Huynh and Nguyen, 2020; Gutiérrez-Romero, 2021; Lv, 2020; Salinas et al., 2021).

As explained above, the objective of this study is to jointly test a group of variables related to each of the three theoretical perspectives that exist about the informal economy. The conceptual framework of the dual view places economic development as the fundamental determinant of the size of the informal economy (La Porta and Shleifer, 2008, 2014). GDP per capita is used in many studies as indicator of economic development (Elgin and Oztunali, 2014; La Porta and Shleifer, 2008). Regarding the legal view, that emphasizes institutional quality as the main determinant of the composition of the formal and informal economy (Dau and Cuervo-Cazurra, 2014; Salinas et al., 2019). We follow Elgin and Solis-Garcia (2015) and Elgin and Oztunali (2014) and use The World Bank Governance Indicators to create an index of institutional quality that combines two aspects of the institutional environment: regulatory quality and rule of law. Finally, the structuralist view highlights trade openness as one of the most relevant causes of the emergence of subcontracting and outsourcing strategies, which in turn are related to a larger size of the informal economy. The data of trade openness has been collected from World Development Indicators. The definition of variables is presented in Table 1.

Table 2 shows the descriptive statistics of the data and the correlation matrix of the variables used in the econometric estimations. Panel A of Table 2 shows that the variables form a balanced panel with 2926 observations, within a period of 22 years ($T = 22$) and the total number of countries in our sample is 133 ($n = 133$). The informal economy variable shows greater variability over time between countries than within countries, the standard deviation (SD) between countries is 1.12 and within countries is 0.03. Similarly, the logGDP variable also shows greater variability between countries than within countries since the between SD of 1.56 is greater than the within SD of 0.20. In addition, the institutional quality variable presents a higher variability across countries than within countries since the between SD of 0.19 is higher than the within SD of 0.04. Finally, we observe that trade openness presents a higher variability across countries than within

Table 1
Description of variables and data sources.

Variable	Symbol	Description	Measure	Data source
Informal economy	IEC	It is the size of the informal economy as a percentage of GDP. Normalized between 0 and 1	Percent of GDP	Medina and Schneider (2018)
Log GDP	LogGDP	Log GDP per capita (in constant 2010)	$USD	World Bank
Institutional quality	INQ	It is a measure that combines the quality of regulation and the strength of a country's rule of law. Normalized between 0 and 1. A value close to 1 indicates a stronger institutional quality.	Index	World Bank Governance Indicators
Trade openness	TRO	Trade is the sum of exports and imports of goods and services measured as a share of gross domestic product. Normalized between 0 and 1.	Percent of GDP	World Bank

Table 2
Descriptive statistics and correlation matrix of the variables.

	IEC	logGDP	INQ	TRO
Panel A: Descriptive statistics				
Mean	0.30	8.48	0.56	0.85
Std. Dev. (Overall)	0.13	1.56	0.19	0.54
Std. Dev. (Between)	0.12	1.56	0.19	0.52
Std. Dev. (Within)	0.03	0.20	0.04	0.16
Min	0.05	5.36	0.08	0.15
Max	0.70	11.62	0.96	4.42
N	2,926	2,926	2,926	2,926
n	133	133	133	133
T	22	22	22	22
Panel B: Correlation matrix				
IEC	1			
	–			
logGDP	−0.709***	1		
	(0.000)	–		
INQ	−0.726***	0.731***	1	
	(0.000)	(0.000)	–	
TRO	−0.265***	0.331***	0.357***	1
	(0.000)	(0.000)	(0.000)	–

The asterisks in the parameters mean: $^{*}p < 0.05$, $^{**}p < 0.01$, $^{***}p < 0.001$. *p*-value in parenthesis.

Table 3
Multicollinearity test.

Variable	VIF	SQRT VIF	Tolerance	*R*-squared
LogGDP	3.26	1.81	0.3067	0.6933
INQ	3.33	1.82	0.3006	0.6994
TRO	1.15	1.07	0.8686	0.1314
Mean VIF	2.58			

countries. On the other hand, panel B of Table 2 shows the correlation matrix between the variables, where we can observe that every independent variable is negatively correlated with the size of informal economy.

Because the correlation matrix evidenced some correlations higher than 50% between the independent variables, we suspected multicollinearity problems. Therefore, we tested for severe multicollinearity problems in our models using the variance inflation factor (VIF) test. Table 3 shows the VIF values for each variable, where it is evident that the maximum VIF value was 3.26, while the mean VIF was 2.58. These values reveal that multicollinearity is not a problem in our analysis because no VIF value exceeds the upper threshold of 5 (Nam et al., 2014).

4. Methodology

Before starting the econometric analyses, we perform a correlation matrix to observe the direction and significance of the relationships between variables. This matrix helps us to detect high correlation between the independent variables and helps us to eliminate the suspicion of multicollinearity problems (redundant information) in the model. Subsequently,

we performed a more formal test to verify multicollinearity problems. We use the VIF test whose results were shown in the previous section. In this test, it is concluded that multicollinearity problems exist if the VIF exceeds the commonly accepted threshold of 5 and 10 (Cullen et al., 2014).

Subsequently, in order to capture the impact of GDP per capita, institutional quality and trade openness on the size of the informal economy, the basic regression model for this study is given as follows:

$$IEC_{i,t} = \beta_0 + \beta_1 logGDP_{i,t} + \beta_2 INQ_{i,t} + \beta_4 TRO_{i,t} + \varepsilon_{i,t} + u_i \quad (1)$$

where $IEC_{i,t}$ represents the size of the informal economy (as a percentage of GDP) of country i in period t. β_0 represents the constant. Meanwhile, β_1, β_2, β_3 measure the effect of, GDP per capita, institutional quality and trade openness on the informal economy respectively; $\varepsilon_{i,t}$ represents the error term and u_i measures the effect of an unobservable and time invariant term.

The first step in the empirical analysis is to test for dependence between cross-section units. One of the weaknesses of standard econometric approaches to panel data is that they assume that each cross-section is uncorrelated with the others over time. However, these assumptions are difficult to sustain considering that the current globalization and liberalization process has deepened the interconnection between countries, thus generating spillovers. We test for cross-sectional dependence between countries by applying the test developed by Pesaran (2004). Later, we analyze the homogeneity of the slope coefficients to account for country-specific characteristics. Next, we employed the unit root tests developed by Pesaran (2007) to find the order of integration of the series. In the fourth step, to verify the existence of a long-run relationship, we employ a second-generation panel cointegration test developed by Westerlund (2007). Once the presence of the cointegrating relationship is confirmed, the long and short-run coefficients can be estimated using CS-ARDL models (Chudik and Pesaran, 2015). Finally, we estimate the direction of causality using the test proposed by Dumitrescu and Hurlin (2012). All these steps are detailed below.

4.1. Cross-sectional dependence tests

Firstly, we check common shock effects across countries. Therefore, we employed the cross-sectional dependence tests proposed by Pesaran (2004). The test verifies the null hypothesis that there is no cross-sectional dependence against the alternative hypothesis in which the cross-sectional dependence is present. In addition, we used the weak cross-sectional dependence test proposed by Pesaran (2015). This test measures the strength of the cross-sectional dependence of the errors. Cross-sectional dependence tests help us control the fact that due to the high degree of international economic integration, unobserved common shocks tend to propagate regionally and globally, which inevitably increases cross-country dependencies. Which may cause biased results if cross-sectional dependence is not controlled.

4.2. Slope homogeneity test

Second, we employ the slope heterogeneity test for panel data proposed by Pesaran and Yamagata (2008). This test allows us to verify whether the slope coefficients of interest are homogeneous across various cross sections. In general, the slope heterogeneity tests help us control the fact that both GDP per capita, institutional quality and trade are heterogeneous across all countries in our sample, not considering this fact could lead to biased results.

4.3. Panel unit root tests

Subsequently, we employed the second generation Cross-sectional Augmented IPS (CIPS) and Cross-sectional Augmented Dickey–Fuller (CADF) unit root tests proposed by Pesaran (2007). It should be emphasized that first generation unit root tests provide misleading conclusions and spurious results due to false rejection of the null hypothesis and data size problems (Banerjee, 2001). In contrast, second generation unit root tests have the main advantage that these tests take into account cross-sectional dependence and slope heterogeneity. Therefore, we use the second-generation tests to examine whether the series has any order of integration.

4.4. Panel co-integration tests

After exploring the unit root properties of all variables, the next step is to check for the existence of panel cointegration between the underlying variables. We use the error correction based cointegration tests developed by Westerlund (2007). The test allows us to determine the presence of long-run equilibrium or co-movement relationships. In contrast to the first generation cointegration tests, this test considers the heterogeneity and cross-sectional dependence of the error term. As mentioned before, controlling these issues is important in order not to obtain erroneous results about two variables that cointegrate. In addition, the results of this cointegration test is that they are sensitive to the length of the terms in the sample with a short time dimension (Persyn and Westerlund, 2008). Another advantage of Westerlund (2007) tests is that it can accommodate several structural breaks in the panel series and point to the location of the structural break endogenously based on the method of Bai and Perron (2003) if desired. In this test, the null hypothesis is the absence of cointegration. To test for the absence of cointegration, Westerlund (2007) developed four test statistics: Ga, and Gt are the group mean statistics, while Pa and Pt are panel statistics. The first two statistics test for the existence of cointegration for at least one of the cross-sectional units. The second two statistics test for the existence of cointegration for the panel as a whole. This test is based on structural rather than residual dynamics and has good properties in small samples.

4.5. Panel co-integration regressions

Next, the long-run and short-run coefficients are estimated using the CS-ARDL method proposed by Chudik and Pesaran (2015). This method has some advantages compared to other methods. First, the CS-ARDL estimator is that it accounts for slope heterogeneity and cross-section dependence, providing efficient results (Chudik and Pesaran, 2015; Chudik et al., 2016), while estimating the short- and long-term coefficients. Second, the CS-ARDL estimator is capable of addressing both short- and long-run cross-section dependence. Third, the CS-ARDL method can provide robust estimates even when the variables are incorporated by different orders of integration, such as I (0) or I (1). Fourth, this method avoids the endogeneity problem by adding the lagged cross-sectional averages into the model. The CS-ARDL equation can be estimated as follows:

$$\Delta IEC_{it} = \mu_i + \Phi_i \left(IEC_{it-1} - \beta_i X_{it-1} - \varphi_{1i}\overline{IEC}_{t-1} - \varphi_{2i}\overline{X}_{t-1} \right) + \sum_{j=1}^{p-1} \lambda_{ij} \Delta IEC_{it-j} + \sum_{j=0}^{q-1} \zeta_{ij} \Delta X_{it-j} + \eta_{1i}\Delta\overline{IEC}_t + \eta_{2i}\Delta\overline{X}_t + \varepsilon_{it} \quad (2)$$

where ΔIEC_{it} is the dependent variable; X_{it} includes all independent variables during the long run; $\overline{IEC}_{t-1}$ is the mean of the variable for the long run; $\overline{X}_{t-1}$ denotes the mean of independent variables for the long run; ΔIEC_{it-j} represent the dependent variable in the short run; ΔX_{it-j} specifies independent variables during the short run; $\Delta\overline{IEC}_t$ denotes the mean of the dependent variable during the short run; $\Delta\overline{X}_t$ consist of the mean of independent variables during the short run and ε_{it} represents the error term. Where j stands for the cross-sectional dimension and $j = 1 \ldots J$ and t represents time and $t = 1 \ldots T$; β_s signifies the coefficients of the independent variables; λ_{ij} is the short-run coefficient of the dependent variable; ζ_{ij} is the short-run coefficients of the independent variables; η_{1i} represents the mean of dependent variables and η_{2i} denotes the mean of independent variables in the short run.

In addition, we use a wide range of estimation techniques to compare and test the robustness of the model. First, we use estimators commonly referred to as first generation estimators such as fully modified ordinary least squares (FMOLS) and the dynamic ordinary least squares (DOLS) estimator. These estimators do not take into account cross-sectional dependence and slope heterogeneity. However, DOLS and FMOLS take into consideration small sample bias and endogeneity bias by using leads and lags of the first differenced regressors (Sharma et al., 2020). Second, an estimator robust to cross-sectional dependence such as the Augmented Mean Group (AMG) estimator is used. The results of the DOLS and FMOLS models are presented in Table A.1 and the results of the AMG estimator are presented in Table 10.

4.6. Panel causality test

Finally, we employ the panel Granger non-causality technique proposed by Dumitrescu and Hurlin (2012). The test allows us to investigate the causal relationships between variables and to identify the direction of causality of the variables. The conclusion in this test is that there is unidirectional causality, bidirectional causality or no causality. Furthermore, the main advantage of this Dumitrescu and Hurlin (2012) causality test is that it is robust with slope heterogeneity and cross-sectional dependence, and does not require N to be of a size larger than T. Performing causality direction tests helps us to establish the direction of impact from one variable to another. This is especially useful in formulating policy recommendations, since it is known with greater certainty which problem (cause) should be addressed to achieve an effect on an outcome variable. Eq. (3) formalizes the Dumitrescu–Hurlin panel causality test as follows:

$$\Delta IEC_{it} = \alpha_i + \sum_{k=1}^{K} \gamma_i^{(k)} \Delta IEC_{it-k} + \sum_{k=1}^{K} \beta_i^{(k)} \Delta x_{it-k} + \varepsilon_{it} \quad (3)$$

where, α_i represents the constant vector; $\gamma_i^{(k)}$ is the lag parameter; $\beta_i^{(k)}$ is the slope of the coefficient, and k denotes the lag length. To test the null hypothesis that there is no causal relationship in the panel, Dumitrescu and Hurlin calculated the Wald test statistic for all panel by taking the average of the individual Wald statistics.

5. Results and discussion

5.1. Results of cross-sectional dependency

First, in Panel A of Table 4, the results of the Pesaran (2004) cross-sectional dependence test are shown. The results suggest that the null hypothesis of cross-section independence should be rejected for all variables in the model at 1% significance. Therefore, the results show that all variables are highly significant and indicates cross-sectional dependence among sample countries. In other words, the presence of spillover effects on other economies is noted. In addition, we also analyze whether the nature of cross-sectional dependence is weak or strong. Panel B of Table 4 presents the Pesaran (2015) test for weak cross-sectional dependence. This test suggests rejecting the null hypothesis that errors are weakly cross-sectional dependent. Therefore, the cross-sectional dependence is strong.

Table 4
Cross-sectional dependence test.

	Panel A: Cross section dependency tests of Pesaran (2004)				
	Variables	IEC	logGDP	INQ	TRO
Global	test statistic	341.614***	263.241***	2.074**	79.837***
	p-value	0.000	0.000	0.030	0.000
HICs	test statistic	111.414***	81.277***	4.163***	61.641***
	p-value	0.000	0.000	0.000	0.000
HMICs	test statistic	79.298***	77.31***	5.164***	16.811***
	p-value	0.000	0.000	0.000	0.000
LMICs	test statistic	97.541***	88.055***	12.697***	12.838***
	p-value	0.000	0.000	0.000	0.000
LICs	test statistic	51.355***	24.106***	11.132***	12.256***
	p-value	0.000	0.000	0.000	0.000
	Panel B: Test of Pesaran (2015) for weak cross-sectional dependence				
	Variables	IEC	logGDP	INQ	TRO
Global	test statistic	438.269***	439.315***	435.695***	429.238***
	p-value	0.000	0.000	0.000	0.000
HICs	test statistic	140.583***	140.933***	140.656***	139.576***
	p-value	0.000	0.000	0.000	0.000
HMICs	test statistic	97.586***	97.798***	96.775***	95.484***
	p-value	0.000	0.000	0.000	0.000
LMICs	test statistic	120.748***	121.001***	119.594***	118.303***
	p-value	0.000	0.000	0.000	0.000
LICs	test statistic	71.098***	71.258***	70.428***	68.400***
	p-value	0.000	0.000	0.000	0.000

The asterisks in the parameters mean: $^{*}p < 0.05$, $^{**}p < 0.01$, $^{***}p < 0.001$.

Table 5
Slope homogeneity test.

Group	Test	$\tilde{\Delta}$	$\tilde{\Delta}_{adj}$
Global	test statistic	40.088***	45.927***
	p-value	0.000	0.000
HICs	test statistic	18.555***	21.258***
	p-value	0.000	0.000
HMICs	test statistic	20.014***	22.929***
	p-value	0.000	0.000
LMICs	test statistic	21.103***	24.177***
	p-value	0.000	0.000
LICs	test statistic	14.390***	16.485***
	p-value	0.000	0.000

The asterisks in the parameters mean: $^{*}p < 0.05$, $^{**}p < 0.01$, $^{***}p < 0.001$.

5.2. Results of slope homogeneity

Regarding slope heterogeneity, Table 5 shows the results of slope homogeneity tests proposed by Pesaran and Yamagata (2008). The *p*-value results of $\tilde{\Delta}$ and $\tilde{\Delta}_{adj}$ allow us to reject the null hypothesis of slope homogeneity. We consider all these problems in the next steps of our analysis.

5.3. Results of panel unit root test

To test the stationarity of the series, we use the second generation panel unit root test developed by Pesaran (2007) that allows us to check the order of integration of the variables while controlling for cross-sectional dependence and slope heterogeneity. Table 6 presents the results of CIPS and CADF. Both tests reveal that the null hypotheses of a unit root cannot be rejected at their levels, except for HICs and LMICs. Following the literature using the CS-ARDL technique (Usman et al., 2022; Ahmed, 2020; Mehmood, 2022), we have used the variables in the same order of integration I (1), since at this level the null hypothesis of unit root is more strongly rejected.[1] Ahmed (2020) points out that the CS-ARDL estimator provides robust estimates even when the variables are of distinct orders such as I (0) or I (1), and as long as the order of integration is not I (2), the ideal estimation method is the CS-ARDL method, which is the case in our work.

[1] In addition, we have estimated the models for these groups of countries (HICs and LMICs) with these variables in levels (undifferenced) and the results do not change significantly.

Table 6
Results of the second-generation unit root test.

Group	Test	Variables			
		IEC	logGDP	INQ	TRO
Levels					
Global	CIPS	−2.348	−1.912	−1.866	−2.152
	CADF	−2.317	−2.483	−1.917	−2.253
HICs	CIPS	−2.325	−2.001**	−2.384	−1.824
	CADF	−2.328	−2.743**	−2.344	−2.007
HMICs	CIPS	−2.251	−1.669	−2.013	−2.172
	CADF	−2.185	−2.196	−2.234	−2.521
LMICs	CIPS	−2.383	−1.349*	−1.912*	−2.459
	CADF	−2.295	−1.724*	−2.025*	−2.377
LICs	CIPS	−2.563	−2.121	−1.781	−2.333
	CADF	−2.399	−2.514	−1.956	−1.941
First differences					
Global	CIPS	−4.391***	−3.413***	−3.793***	−4.003***
	CADF	−3.189***	−2.782***	−2.924***	−3.095***
HICs	CIPS	−4.286***	−3.055***	−3.909***	−3.549***
	CADF	−3.044***	−2.542***	−3.234***	−2.827***
HMICs	CIPS	−4.492***	−3.677***	−3.524***	−3.968***
	CADF	−3.019***	−3.080***	−3.129***	−2.991***
LMICs	CIPS	−4.603***	−3.660***	−3.775***	−4.125***
	CADF	−3.501***	−2.879***	−2.742***	−3.076***
LICs	CIPS	−4.833***	−4.437***	−3.607***	−4.969***
	CADF	−3.461***	−3.560***	−2.485***	−3.631***

The asterisks in the parameters mean: $^{*}p < 0.05$, $^{**}p < 0.01$, $^{***}p < 0.001$.

Table 7
Result of the Westerlund (2007) cointegration test.

Group	Estadístic	logGDP		INQ		TRO	
		Valor	*p*-value	Valor	*p*-value	Valor	*p*-value
Global	Gt	−3.993	0.000	−4.165	0.000	−3.827	0.000
	Ga	−21.372	0.000	−22.160	0.000	−20.598	0.000
	Pt	−47.021	0.000	−43.530	0.000	−42.936	0.000
	Pa	−22.407	0.000	−21.776	0.000	−20.807	0.000
HICs	Gt	−3.722	0.000	−4.830	0.000	−3.800	0.000
	Ga	−20.358	0.000	−22.074	0.000	−20.382	0.000
	Pt	−26.020	0.000	−30.308	0.000	−24.374	0.000
	Pa	−21.542	0.000	−21.047	0.000	−19.450	0.000
UMICs	Gt	−3.778	0.000	−3.673	0.000	−3.822	0.000
	Ga	−20.887	0.000	−19.894	0.000	−19.374	0.000
	Pt	−22.919	0.000	−21.739	0.000	−23.100	0.000
	Pa	−21.246	0.000	−19.960	0.000	−19.306	0.000
LMICs	Gt	−4.344	0.000	−4.025	0.000	−3.908	0.000
	Ga	−20.903	0.000	−22.147	0.000	−21.158	0.000
	Pt	−22.953	0.000	−20.741	0.000	−20.378	0.000
	Pa	−21.134	0.000	−21.086	0.000	−21.840	0.000
LICs	Gt	−4.417	0.000	−3.856	0.000	−3.776	0.000
	Ga	−24.854	0.000	−26.163	0.000	−22.322	0.000
	Pt	−21.100	0.000	−17.724	0.000	−17.362	0.000
	Pa	−26.251	0.000	−26.305	0.000	−22.634	0.000

The *p*-value was obtained through the bootstrapping method with 200 repetitions.

5.4. Panel cointegration results

Next, we verify the potential existence of a long-run relationship between the informal economy and its determinants. To do so, we employ cointegration method of Westerlund (2007). The results presented in Table 7 reject the null hypothesis of no cointegration and reveal a long-run cointegration relationship between the size of the informal economy and the other independent variables, both at the aggregate level and for the different groups of countries.

Table 8
Results of CS-ARDL estimator.

	CS-ARDL				
	GLOBAL	HICs	UMICs	LMICs	LICs
Short run estimates					
logGDP	−0.154***	−0.187***	−0.188***	−0.108***	−0.128**
	(0.016)	(0.019)	(0.033)	(0.033)	(0.048)
INQ	−0.152***	−0.130***	−0.207***	−0.177***	−0.014
	(0.016)	(0.021)	(0.037)	(0.030)	(0.052)
TRO	−0.072***	−0.055***	−0.094***	−0.083***	−0.071***
	(0.009)	(0.011)	(0.018)	(0.027)	(0.020)
Long run estimates					
logGDP	−1.154***	−1.187***	−1.188***	−1.108***	−1.128***
	(0.016)	(0.019)	(0.033)	(0.033)	(0.048)
INQ	−0.139***	−0.113***	−0.182***	−0.168***	−0.027*
	(0.015)	(0.016)	(0.032)	(0.029)	(0.013)
TRO	−0.067***	−0.048***	−0.082***	−0.082**	−0.068***
	(0.010)	(0.010)	(0.016)	(0.031)	(0.018)
Observations	2,793	924	777	630	462
N	133	44	37	30	22

The dependent variable is informal economy. The asterisks in the parameters mean: $^{*}p < 0.05$, $^{**}p < 0.01$, $^{***}p < 0.001$. Standard errors in parenthesis.

5.5. Results of panel co-integration regressions

Due to the presence of cross-sectional dependency and slope heterogeneity in our models, this study employs the CS-ARDL model proposed by Chudik and Pesaran (2015) to estimate the short- and long-run coefficients. By applying this estimator, we solved cross-sectional dependency both for short and long-run. The results of the models using CS-ARDL are shown in Table 8.

Our first hypothesis suggests a negative relationship between the level of economic development and the size of the informal economy. Indeed, this is evident at the global level and by country group, both in the long and short run. Thus, our first hypothesis related to the dual view of informality is accepted. These findings are in line with La Porta and Shleifer (2008), Elgin and Oztunali (2014) and Zhanabekov (2022), who argue that as GDP per capita increases, the size of the informal economy decreases. At the country level, various works find that GDP per capita affect negatively the size of the informal economy (Bologna, 2016; Dell'Anno et al., 2007; Garcia and de Freitas Sanches, 2021). Recently, Tran (2021) examined seven Southeast Asian Nations and found that an increase of GDP per capita results in a lower informal economy. The same conclusion was made by Baklouti and Boujelbene (2020) for the case of 17 Middle East and North Africa (MENA) countries, and by Akonor (2020) for 16 West African countries. Lastly, Ginevicius et al. (2020) and Navickas et al. (2019) found the same results for several EU member states.

Our second hypothesis corresponds to the legal view. This perspective argues about a negative relationship between institutional quality and the size of the informal economy. The hypothesis is verified in both the short and the long run at the global level and by country group. However, the short-run coefficient appears to be insignificant for the case of the group of low-income countries. These results are supported by Feige (1990), Perry and Maloney (2007), Loayza and Rigolini (2011), Hassan and Schneider (2016), Saunoris and Sajny (2017), Berdiev and Saunoris (2016), Salinas et al. (2020). However, these papers have used traditional panel data methods with potential biases in their results.

Our third hypothesis corresponds to the structuralist view and argues that openness to trade would cause a rise in the size of informal economy. The evidence we obtain from our analysis challenges this perspective. The results show that the coefficient of trade openness is negative and significant in both the short and long run. This relationship holds both at the global level and by country group. It must be noted that even in the case of low and lower-middle income countries, where the structuralist view emphasizes the use of outsourcing mechanisms by firms to protect themselves from international competition, this hypothesis does not hold.

Previous empirical studies have generated mixed evidence and inconclusive results on the relationship between trade openness and the size of the informal economy (see Ghosh and Paul, 2008; Birinci, 2013; Dada et al., 2022). These papers have used various measures of the informal economy, different econometric strategies, and some of them have looked at specific countries. As indicated above, our econometric strategy allows us to overlook many shortcomings of traditional econometric approaches and provide results that support the empirical literature that argues about a negative relationship between the informal economy and trade openness for a large sample of countries (Paz, 2014; Blanton et al., 2018; Canh et al., 2021; Zhanabekov, 2022). Moreover, these results are congruent with Goldberg and Pavcnik (2003) for Latin American countries, and Birinci (2013) for 12 advanced economies.

Since previous studies have suggested the existence of a nonlinear relationship between the size of informal economy and GDP per capita (Wu and Schneider, 2019), we address this concern by including the square value of GDP per capita

Table 9
U-shape relationship with CS-ARDL estimator.

	CS-ARDL GLOBAL
Short run estimates	
logGDP	−0.247***
	(0.029)
$logGDP^2$	1.259*
	(0.553)
INQ	−0.152
	(0.118)
TRO	−0.068***
	(0.009)
Long run estimates	
logGDP	−1.247***
	(0.029)
$logGDP^2$	0.181**
	(0.501)
INQ	−0.132***
	(0.016)
TRO	−0.060***
	(0.008)
Observations	2,793
N	133

The dependent variable is informal economy. The asterisks in the parameters mean: $^{*}p < 0.05$, $^{**}p < 0.01$, $^{***}p < 0.001$. Standard errors in parenthesis.

Table 10
Robust analysis with Augmented Mean Group (AMG) estimator.

	GLOBAL	HICs	UMICs	LMICs	LICs
logGDP	−0.112***	−0.117***	−0.105***	−0.089**	−0.087*
	(0.011)	(0.012)	(0.019)	(0.029)	(0.039)
INQ	−0.124***	−0.083***	−0.195***	−0.161***	−0.068
	(0.014)	(0.013)	(0.036)	(0.032)	(0.045)
TRO	−0.042***	−0.014*	−0.050***	−0.064**	−0.062***
	(0.008)	(0.004)	(0.014)	(0.020)	(0.016)
cdp	0.916***	0.984***	0.863***	0.839***	0.960***
	(0.057)	(0.071)	(0.105)	(0.129)	(0.089)
trend	−0.000*	−0.000	−0.000	−0.000	−0.000
	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)
Constant	−0.005***	0.000	0.005***	−0.006**	−0.000
	(0.001)	(0.000)	(0.001)	(0.002)	(0.001)
Observations	2,793	924	777	630	462
N	133	44	37	30	22

The dependent variable is informal economy. The asterisks in the parameters mean: $^{*}p < 0.05$, $^{**}p < 0.01$, $^{***}p < 0.001$. Standard errors in parenthesis. Variable cdp refers to the common dynamic process. Variable trend refers to the group-specific linear trend terms.

($logGDP^2$) to observe whether this type of relationship does indeed exist. In Table 9 we present the results only for the global sample. It should be noted that dividing the sample by country groups limits our conclusions because within the groups there is little heterogeneity among countries. The results in Table 9 support the existence of the U-shaped relationship between the size of the informal economy and GDP per capita at the global level. This finding is in line with Wu and Schneider (2019) who evidenced a U-shaped relationship over the period from 1996 to 2015 for 158 countries. However, they used traditional panel regression methods with potential biases discussed previously.

In order to verify our findings, we perform some alternative estimation techniques. Initially, we implement the FMOLS and DOLS estimator. The outcomes of FMOLS and DOLS are reported in Table A.1. Overall, the long-run FMOLS and DOLS coefficients confirm the findings that we have previously discussed using CS-ARDL. Additionally, we apply a more efficient estimator such as the Augmented Mean Group (AMG) estimator. The coefficients derived from the AMG estimator are robust and reliable, as they account for cross-sectional dependence and allow for heterogeneous slope coefficients across panel observations (Paramati et al., 2017). We present the results of the AMG estimator in Table 10. Again, the results are found to be consistent with the estimates of CS-ARDL estimator.

Table 11
Results of Dumitrescu and Hurlin (2012) panel causality test.

Null hypothesis	Group	Z-bar	*p*-value	Result
IEC≠>logGDP	Global	3.5875	0.0003	Causality relationship
	HICs	4.2065	0.0000	Causality relationship
	UMICs	2.6641	0.0077	Causality relationship
	LMICs	−0.2114	0.8326	No causality relationship
	LICs	−0.3362	0.7367	No causality relationship
logGDP ≠> IEC	Global	2.7813	0.0054	Causality relationship
	HICs	0.2125	0.0317	Causality relationship
	UMICs	10.0090	0.0000	Causality relationship
	LMICs	2.4293	0.0151	Causality relationship
	LICs	0.7723	0.0399	Causality relationship
IEC ≠> INQ	Global	9.2223	0.0000	Causality relationship
	HICs	9.6916	0.0000	Causality relationship
	UMICs	−2.2473	0.0246	Causality relationship
	LMICs	−0.8721	0.3831	No causality relationship
	LICs	3.0069	0.0026	Causality relationship
INQ ≠> IEC	Global	18.5906	0.0000	Causality relationship
	HICs	9.4063	0.0000	Causality relationship
	UMICs	11.6713	0.0000	Causality relationship
	LMICs	13.7207	0.0000	Causality relationship
	LICs	0.3082	0.0500	Causality relationship
IEC ≠> TRO	Global	2.8123	0.0049	Causality relationship
	HICs	3.6219	0.0003	Causality relationship
	UMICs	1.2869	0.0281	Causality relationship
	LMICs	10.1198	0.0000	Causality relationship
	LICs	3.0052	0.0027	Causality relationship
TRO ≠> IEC	Global	0.9199	0.0357	Causality relationship
	HICs	15.0031	0.0000	Causality relationship
	UMICs	1.7164	0.0261	Causality relationship
	LMICs	5.7065	0.0000	Causality relationship
	LICs	5.4894	0.0000	Causality relationship

Note: ≠> means variable *x* does not Granger cause variable.

5.6. Results of panel causality test

Finally, in the last step of this research and after estimating the short and long-run CS-ARDL coefficients, we apply the method of Dumitrescu and Hurlin (2012) to investigate the causal relationships between the variables. The results of the heterogeneous panel causality test of Dumitrescu and Hurlin (2012) are presented in Table 11. Among the most relevant results of these bivariate causality tests, we find that there is a bidirectional causal relationship between the informal economy and the GDP per capita for the global, high income and upper middle- income panels, while there is a causal relationship running from the GDP per capita towards the informal economy for lower middle-income and low-income countries. On the other hand, there is a bidirectional causality between the informal economy and institutional quality at the global, high-income, upper middle-income and lower middle-income panels, while we observe a unidirectional causality running from institutional quality to the informal economy in low-income countries. Finally, we find bidirectional causality between trade openness and the informal economy at the global panel and by groups of countries. In addition, Table A.2 in Appendix shows the causality tests between independent variables. Overall, our findings are partially in line with Salinas et al. (2021) who found causality in the same direction between GDP per capita and the informal economy and also bidirectional causality between institutional quality and the informal economy. These findings indicate that public policies that focus on promoting growth, institutional quality and international trade could have a direct and negative impact on the informal economy.

6. Conclusions and policy implications

Our main objective in this study is to examine the short and long run and causal relationships between informal economy and three independent variables, specifically GDP per capita, institutional quality and trade openness, each of which associated with the three views of the informal economy that were previously discussed. Our study considers 133 countries over 1995–2017. In addition, we considered four sub-panels related to the income level of the countries in the sample (groups of high, middle, lower middle- and low-income countries). To fulfill this objective, we used second-generation panel econometrics tools, which is suitable for heterogeneous panel and accounts for the presence of cross-section dependence. Also, we apply CS-ARDL model to estimate the short and long-run relationships between our variables of interest. This study contributes to the existing literature by jointly examining three views of the informal economy using advanced second-generation econometric techniques.

We investigate three main hypotheses. The first hypothesis refers to the dual view, which suggest a negative relationship between GDP per capita and the size of the informal economy. We find both long and short-run relationships between the size of the informal economy and GDP per capita. Therefore, this hypothesis is confirmed, and it is expected that as countries reach higher levels of development, the size of the informal economy will decrease. This relationship was found both at the global and country group level—. However, the results derived from the hypothesis1B, suggest that the relationship between GDP per capita and the size of the informal economy is non-linear for the entire sample. Then, the share of the informal sector decreases as countries reach higher levels of development but tends to increase again as countries achieve a given threshold of development. This occurs specifically for those countries in the highest stage of development. Thus, the results provide support for the existence of a U-shaped relationship between GDP per capita and the size of the informal economy. This may be due to the fact that new technologies are altering labor relations, and this is more evident in the case of countries located at the technological frontier. Basically, technological change is creating new forms of labor relations that can lead to informality. These findings suggest a reinterpretation of earlier findings that have emphasized a linear relationship between GDP per capita and the size of the informal economy.

The second hypothesis is related to the legal vision, which argues about a negative relationship between the institutional quality of a country and the size of the informal economy. This argument is also verified by our findings. The empirical results of CS-ARDL show that there is a negative and statistically significant relationship between institutional quality and the size of the informal economy both in the long and short-run. These results hold both at the global and country group level, except for LICs, where there is no significant relationship between the variables. Thus, as countries improve their regulatory and legal framework, the size of their informal sector will tend to fall.

Regarding the third hypothesis and in line with the structuralist view, we hypothesize that the higher the level of trade openness the larger the size of the informal economy. However, our results challenged this perspective. In fact, we find a negative and significant relationship between trade openness and the informal economy both at the global level and by country group. This finding is relevant, since several theoretical models predict a positive relationship, arguing that there is a symbiotic relationship between the formal and informal economy, where some production processes are outsourced to small informal firms (Jones et al., 2006). Then, our results suggest that as countries become increasingly integrated into the global economy, they reduce their level of informal activity, probably because informal activities such as smuggling for tariff evasion are reduced (Blanton et al., 2018). In addition, international competitors and multinational corporations may disseminate to local firms the best regulatory standards, norms, and industrial practice and thus encourage participation in the formal sector (Greenhill et al., 2009; Adolph et al., 2017).

The empirical findings of our study provide several important policy implications. For instance, CS-ARDL estimator and Dumitrescu–Hurlin panel causality test demonstrate the effectiveness of GDP per capita, institutional quality and trade openness in diminishing the size of the informal economy. Therefore, the empirical findings suggest that governments have at their disposal several instruments, and thus notable leeway for informality reduction. Therefore, the primary policy implication is that the success of any reform will depend to a great extent on the government's ability to simultaneously implement a mix of policies aimed at stimulating economic growth, reduce their regulatory burden, improve the legal framework, and open the economy to international trade.

Nevertheless, based on our results of a U-shaped relationship between informality and GDP per capita, public policy should be designed conditional on the level of development of a certain country. Specifically, policy makers in developed countries should not assume an inevitable decline in informal activities, in the presence of a sustainable economic growth rate. On the contrary, technological change seems to be blurring the difference between the formal and informal economy in some sectors, and therefore it is essential to adapt and redefine the institutional framework according to the new technological revolution.

Finally, we must acknowledge the limitations of the work. For instance, the work has not addressed the possible-or-not substitution between formal and informal activity. Specifically, the results do not allow us to infer whether merely by improving the legal and regulatory framework there will be a massive conversion of informal firms to formal status just as suggested by De Soto (1989). Therefore, future research on this topic is necessary.

CRediT authorship contribution statement

Aldo Salinas: Supervision, Validation, Conceptualization, Formal analysis, Methodology, Writing – original draft, Writing – review & editing. **Cristian Ortiz:** Data curation, Formal analysis, Methodology, Software, Writing – original draft. **Javier Changoluisa:** Supervision, Formal Analysis, Conceptualization, Writing – original draft. **Moreno Muffatto:** Supervision, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix

See Tables A.1 and A.2.

Table A.1
Results of base line using FMOLS and DOLS estimator.

	FMOLS					DOLS				
	GLOBAL	HICs	UMICs	LMICs	LICs	GLOBAL	HICs	UMICs	LMICs	LICs
logGDP	−0.12***	−0.10***	−0.15***	−0.14***	−0.10***	−0.11***	−0.06***	−0.11***	−0.17***	−0.10***
	(−54.58)	(−31.30)	(−28.40)	(−32.21)	(−15.48)	(−24.40)	(−12.87)	(−12.46)	(−16.63)	(−6.22)
INQ	−0.12***	−0.11***	−0.16***	−0.14***	−0.07***	−0.08***	−0.10***	−0.01	−0.11***	−0.16***
	(−23.54)	(−15.86)	(−11.99)	(−13.52)	(−4.11)	(−3.03)	(−1.77)	(−0.42)	(−2.61)	(−2.44)
TRO	−0.09***	−0.07***	−0.10***	−0.10***	−0.08***	−0.08***	−0.06***	−0.07***	−0.16***	−0.02
	(−50.22)	(−30.04)	(−21.08)	(−33.11)	(−15.01)	(−13.62)	(−9.29)	(−6.42)	(−10.45)	(0.18)
Observations	2,793	924	777	630	462	2,793	924	777	630	462
N	133	44	37	30	22	133	44	37	30	22

The dependent variable is informal economy. The asterisks in the parameters mean: $^{*}p < 0.05$, $^{**}p < 0.01$, $^{***}p < 0.001$. *t*-statistic in parenthesis.

Table A.2
Results of Dumitrescu and Hurlin (2012) panel causality test for independents variables.

Null hypothesis	Group	Z-bar	*p*-value	Result
logGDP ≠> *TRO*	Global	5.4433	0.0000	Causality relationship
	HICs	10.6737	0.0000	Causality relationship
	UMICs	3.2778	0.0010	Causality relationship
	LMICs	1.5219	0.1280	No causality relationship
	LICs	1.0651	0.2869	No causality relationship
TRO ≠>*logGDP*	Global	16.1742	0.0000	Causality relationship
	HICs	13.9238	0.0000	Causality relationship
	UMICs	1.5635	0.1179	No causality relationship
	LMICs	0.9845	0.3249	No causality relationship
	LICs	1.8329	0.0668	No causality relationship
logGDP ≠> *INQ*	Global	9.6821	0.0000	Causality relationship
	HICs	0.3613	0.0179	Causality relationship
	UMICs	7.0868	0.0000	Causality relationship
	LMICs	5.4647	0.0000	Causality relationship
	LICs	4.5798	0.0000	Causality relationship
INQ ≠> *logGDP*	Global	13.9039	0.0000	Causality relationship
	HICs	5.4591	0.0000	Causality relationship
	UMICs	10.9049	0.0000	Causality relationship
	LMICs	12.0058	0.0000	Causality relationship
	LICs	0.9024	0.0268	Causality relationship
TRO ≠> *INQ*	Global	11.3079	0.0000	Causality relationship
	HICs	9.3991	0.0000	Causality relationship
	UMICs	0.2728	0.0450	Causality relationship
	LMICs	6.0570	0.0000	Causality relationship
	LICs	4.1539	0.0000	Causality relationship
INQ ≠> *TRO*	Global	16.5703	0.0000	Causality relationship
	HICs	9.2034	0.0000	Causality relationship
	UMICs	12.8994	0.0000	Causality relationship
	LMICs	8.2019	0.0000	Causality relationship
	LICs	2.7141	0.0066	Causality relationship

Note: ≠> means variable *x* does not Granger cause variable.

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